

MIT integration bee - 2018

Notación y Consideraciones Generales

A lo largo del documento vamos a utilizar la siguiente notación, en algunos problemas más complejos utilizaremos la notación usual:

$$\{ f(x) \}_a^b := \int_a^b f(x) dx, \quad [f(x)] := \frac{d}{dx} f(x)$$

Recursos obtenidos de: <https://math.mit.edu/~yyao1/integrationbee.html>

Herramientas que te pueden ser útiles:

- Calculadora gráfica: <https://www.desmos.com/calculator?lang=es>
- Calculadora de integrales: <https://mathdf.com/es/>
- Calculadora de integrales: <https://www.wolframalpha.com/>
- Motor de búsqueda para fórmulas en latex <https://approach0.xyz/search/>
- Te ayuda con ideas: <https://chat.deepseek.com>

Ejercicio 1

1.
$$\int \frac{e^x}{e^x + 2} dx$$

— — — Demostración — — —

$$I = \ln |e^x + 2| + C \quad (1)$$

- (1): $[\ln(f(x))] = \frac{f'(x)}{f(x)}$

Ejercicio 2

$$\int \sqrt{x \cdot \sqrt[3]{x \cdot \sqrt[4]{x \cdot \sqrt[5]{x \cdots}}}} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ x^{\frac{1}{2}} \cdot x^{\frac{1}{2 \cdot 3}} \cdot x^{\frac{1}{2 \cdot 3 \cdot 4}} \cdots \right\} \\ &= \left\{ x^{\frac{1}{2!}} \cdot x^{\frac{1}{3!}} \cdot x^{\frac{1}{4!}} \cdots \right\} \\ &= \left\{ x^{\sum_{n \geq 2} \frac{1}{n!}} \right\} \\ &= \left\{ x^{\sum_{n \geq 0} \frac{1}{n!} - 2} \right\} \\ &= \left\{ x^{e-2} \right\} \\ &= \frac{x^{e-1}}{e-1} + C \end{aligned} \quad (1)$$

- (1): $e^x = \sum_{n \geq 0} \frac{x^n}{n!}$

Ejercicio 3

$$\int_0^{2018\pi} |\sin(2018x)| dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ |\sin(\alpha x)| \right\}_0^{\alpha\pi} \\ &= \frac{1}{\alpha} \left\{ |\sin(u)| \right\}_0^{\alpha^2\pi} \quad (1) \\ &= \frac{2\alpha^2}{\alpha} \left\{ |\sin(u)| \right\}_0^{\pi/2} \quad (2) \\ &= 2\alpha \left\{ \sin(u) \right\}_0^{\pi/2} \\ &= 2\alpha \left(-\cos(u) \right)_0^{\pi/2} \\ &= 2\alpha \left(-\cos(u) \right)_0^{\pi/2} \\ &= 2\alpha \end{aligned}$$

- (1): $\begin{cases} u &= \alpha x \\ du &= \alpha dx \end{cases}$

- (2) función de periodo π

Ejercicio 4

$$\int \frac{dx}{\tan x + \cot x}$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ \frac{\sin(x) \cos(x)}{\sin^2(x) + \cos^2(x)} \right\} \\ &= \left\{ \sin(x) \cos(x) \right\} \quad (1) \\ &= \frac{\sin^2(x)}{2} + C \end{aligned}$$

- (1): $\cos^2(x) + \sin^2(x) = 1$

Ejercicio 5

$$\int \frac{x^5}{2 + x^{12}} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ \frac{x^5}{2 + (x^6)^2} \right\} \\ &= \frac{1}{6} \left\{ \frac{1}{2 + u^2} \right\} \quad (1) \\ &= \frac{1}{12} \left\{ \frac{1}{1 + \left(\frac{u}{\sqrt{2}}\right)^2} \right\} \\ &= \frac{\sqrt{2}}{12} \arctan \left(\frac{u}{\sqrt{2}} \right) + C \end{aligned}$$

- (1): $\begin{cases} u &= x^6 \\ du &= 6x^5 dx \end{cases}$

Ejercicio 6

$$\int (\cos x \cosh x + \sin x \sinh x) dx$$

— — — Demostración — — —

$$I = \sin(x) \cosh(x) + C \quad (1)$$

- (1): $[\sin(x) \cosh(x)] = \cos(x) \cosh(x) + \sin(x) \sinh(x)$

Ejercicio 7

$$\int \frac{e^x + \cos x}{e^x + \sin x} dx$$

— — — Demostración — — —

$$I = \ln |e^x + \sin(x)| + C \quad (1)$$

- (1): $[\ln(f(x))] = \frac{f'(x)}{f(x)}$

Ejercicio 8

$$\int \sin(\cos(\sin x)) \sin(\sin x) \cos x dx$$

— — — Demostración — — —

$$I = \{ \sin(\cos(u)) \sin(u) \} \quad (1)$$

$$= - \{ \sin(t) \} \quad (2)$$

$$= \cos(t) + C$$

$$= \cos(\cos(\sin(x))) + C$$

$$\bullet (1): \begin{cases} u &= \sin(x) \\ du &= \cos(x) dx \end{cases}$$

$$\bullet (2): \begin{cases} t &= \cos(u) \\ dt &= -\sin(u) du \end{cases}$$

Ejercicio 9

$$\int \frac{dx}{1 + \sin(x)}$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ \frac{\sec(x)}{\sec(x) + \tan(x)} \right\} \\ &= \left\{ \frac{\sec(x)(\sec(x) + \tan(x))}{(\sec(x) + \tan(x))^2} \right\} \\ &= \left\{ \frac{1}{u^2} \right\} \\ &= -\frac{1}{u} + C \end{aligned} \quad (1)$$

$$\bullet (1): \begin{cases} u &= \sec(x) + \tan(x) \\ du &= \sec(x)(\sec(x) + \tan(x)) dx \end{cases}$$

Ejercicio 10

$$\int \frac{\cos x}{1 - \cos(2x)} dx$$

— — — Demostración — — —

$$I = \frac{1}{2} \left\{ \frac{\cos x}{\sin^2(x)} \right\} \quad (1)$$

$$= -\frac{1}{2} \frac{1}{\sin(x)} + C \quad (2)$$

$$= -\frac{\csc(x)}{2} + C$$

$$\bullet (1): \cos(2x) = 1 - 2\sin^2(x)$$

$$\bullet (2): \left[\frac{1}{f(x)} \right] = -\frac{f'(x)}{f^2(x)}$$

Ejercicio 11

$$\int e^x (1/x + \log x) dx$$

— — — Demostración — — —

$$I = e^x \ln(x) + C \quad (1)$$

$$\bullet (1): [e^x f(x)] = e^x (f(x) + f'(x))$$

Ejercicio 12

$$\int \tanh^2(x) dx$$

— — — Demostración — — —

$$I = \{ 1 - \operatorname{sech}^2(x) \} \quad (1)$$

$$I = x - \tanh(x) + C \quad (2)$$

$$\bullet (1): 1 - \tanh^2(x) = \operatorname{sech}^2(x)$$

$$\bullet (2): [\tanh(x)] = \operatorname{sech}^2(x)$$

Nota:

1. Propiedades básicas de las funciones trigonometricas hiperbólicas:

$$\left\{ \begin{array}{ll} \cosh^2(x) - \sinh^2(x) &= 1 \\ 2 \sinh^2(x) &= \cosh(2x) - 1 \\ 2 \cosh^2(x) &= \cosh(2x) + 1 \\ \sinh^2(x) + \cosh^2(x) &= \cosh(2x) \\ 2 \sinh(x) \cosh(x) &= \sinh(2x) \\ (\sinh(x) + \cosh(x))^2 &= \cosh(2x) + \sinh(2x) \end{array} \right.$$

2. Derivadas de las funciones trigonométricas hiperbólicas:

$$\begin{cases} [\sinh(x)] = \cosh(x) \\ [\cosh(x)] = \sinh(x) \\ [\tanh(x)] = \operatorname{sech}^2(x) \\ [\coth(x)] = -\operatorname{csch}^2(x) \\ [\operatorname{sech}(x)] = -\operatorname{sech}(x)\tanh(x) \\ [\operatorname{csch}(x)] = -\operatorname{csch}(x)\coth(x) \end{cases}$$

Ejercicio 13

$$\int \frac{2017x^{2016} + 2018x^{2017}}{1 + x^{4034} + 2x^{4035} + x^{4036}} dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{\alpha x^{\alpha-1} + (\alpha+1)x^{\alpha}}{1 + x^{2\alpha} + 2x^{2\alpha+1} + x^{2\alpha+2}} \right\} \\ &= \left\{ \frac{\alpha x^{\alpha-1} + (\alpha+1)x^{\alpha}}{1 + (x^{\alpha+1} + x^{\alpha})^2} \right\} \end{aligned} \quad (1)$$

$$\begin{aligned} &= \left\{ \frac{1}{1 + u^2} \right\} \quad (2) \\ &= \arctan(u) + C \end{aligned}$$

- (1): Triángulo de Pascal

$$\bullet (2): \begin{cases} u = x^{\alpha+1} + x^{\alpha} \\ du = ((\alpha+1)x^{\alpha} + \alpha x^{\alpha-1}) dx \end{cases}$$

Ejercicio 14

$$\int \frac{\sin(2x) - \sin^2 x}{\cos(2x) - \cos^2 x} dx$$

--- Demostración ---

$$I = \left\{ \frac{\sin(2x) - \sin^2 x}{\cos^2 x - 1} \right\} \quad (1)$$

$$= - \left\{ \frac{\sin(2x) - \sin^2 x}{\sin^2(x)} \right\} \quad (2)$$

$$\begin{aligned} &= - \left\{ \frac{\sin(2x)}{\sin^2(x)} - 1 \right\} \\ &= - \left\{ \frac{2 \sin(x) \cos(x)}{\sin^2(x)} - 1 \right\} \end{aligned} \quad (3)$$

$$\begin{aligned} &= - \{ 2 \cot(x) - 1 \} \\ &= - (2 \ln |\sin(x)| - x) + C \\ &= x - 2 \ln |\sin(x)| + C \end{aligned}$$

$$\bullet (1): \cos(2x) = 2 \cos^2(x) - 1$$

$$\bullet (2): \cos^2(x) + \sin^2(x) = 1$$

$$\bullet (3): \sin(2x) = 2 \sin(x) \cos(x)$$

Ejercicio 15

$$\int \frac{dx}{x^{\frac{25}{25}} \cdot x^{\frac{16}{25}} + x^{\frac{9}{25}}}$$

--- Demostración ---

$$\begin{aligned} I &= 25 \left\{ \frac{t^{16}}{t^{32} + 1} \right\} & (1) \\ &= 25 \left\{ \frac{t^{15}}{(t^{16})^2 + 1} \right\} \\ &= \frac{25}{16} \left\{ \frac{1}{u^2 + 1} \right\} & (2) \\ &= \frac{25}{16} \arctan(u) + C \end{aligned}$$

$$\bullet (1): \begin{cases} x &= t^{25} \\ dx &= 25t^{24} dt \end{cases}$$

$$\bullet (2): \begin{cases} u &= t^{16} \\ du &= 16t^{15} dt \end{cases}$$

Ejercicio 16

$$\int_0^{\pi/2} \frac{\cos(x)}{2 - \cos^2(x)} dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{\cos(x)}{1 + \sin^2(x)} \right\}_0^{\pi/2} & (1) \\ &= \left\{ \frac{1}{1 + u^2} \right\}_0^1 & (2) \\ &= (\arctan(u))_0^1 \\ &= \frac{\pi}{4} \end{aligned}$$

$$\bullet (1): \cos^2(x) + \sin^2(x) = 1$$

$$\bullet (2): \begin{cases} u &= \sin(x) \\ du &= \cos(x) dx \end{cases}$$

Ejercicio 17

$$\int \frac{dx}{(1 + x^2)^{3/2}}$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{\sec^2(\theta)}{\sec^3(\theta)} \right\} & (1) \\ &= \{ \cos(\theta) \} \\ &= \sin(\theta) + C \\ &= \frac{x}{\sqrt{x^2 + 1}} + C \end{aligned}$$

$$\bullet (1): \begin{cases} x &= \tan(\theta) \\ dx &= \sec^2(\theta) d\theta \end{cases}$$

Ejercicio 18

$$\int \frac{dx}{\sqrt{x}\sqrt{x-x^2}}$$

--- Demostración ---

$$\begin{aligned} I &= 2 \left\{ \frac{t}{\sqrt{t^3-t^4}} \right\} & (1) \\ &= 2 \left\{ \frac{1}{\sqrt{t-t^2}} \right\} \\ &= 4 \left\{ \frac{u}{\sqrt{u^2-u^4}} \right\} & (2) \\ &= 4 \left\{ \frac{1}{\sqrt{1-u^2}} \right\} \\ &= 4 \arcsin(u) + C \\ &= 4 \arcsin(x^{1/4}) + C \end{aligned}$$

- (1): $\begin{cases} x = t^2 \\ dx = 2t dt \end{cases}$

- (2): $\begin{cases} t = u^2 \\ dt = 2u du \end{cases}$

Ejercicio 19

$$\int \frac{x-1}{x+x^2 \log x} dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{u-1}{u^2-u \ln u} \right\} & (1) \\ &= \left\{ \frac{1-\frac{1}{u}}{u-\ln u} \right\} \\ &= \ln |u-\ln(u)| + C \end{aligned}$$

- (1): $\begin{cases} x = \frac{1}{u} \\ dx = -\frac{1}{u^2} du \end{cases}$

- (2): $[\ln(f(x))] = \frac{f'(x)}{f(x)}$

Ejercicio 20

$$\int \csc(x) \sec(x) dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{\sin^2(x) + \cos^2(x)}{\sin(x) \cos(x)} \right\} & (1) \\ &= \{ \tan(x) + \cot(x) \} \\ &= -\ln |\cos(x)| + \ln |\sin(x)| + C \end{aligned}$$

- (1): $\cos^2(x) + \sin^2(x) = 1$

